VB

ASSEMBLYAI VOICE AGENT HACKATHON

VoiceBridge AI

Operate the web with your voice.
A voice-first accessibility companion — not a chat wrapper.

Built on AssemblyAI Voice Agent API · Multi-engine fallback · Production security

THE PROBLEM

Pixel-first UIs exclude people who need them most

- Blind, low-vision, and motor-impaired users fight mouse-first dashboards
- Most voice UIs are chat wrappers — they talk, they do not operate a product
- When cloud STT fails, voice apps go dark for the people who need them most

1B+
People with disabilities worldwide

2.2B
Vision impairment (WHO)

WCAG
Legal mandates expanding

THE SOLUTION

A dashboard you can operate with speech

1 Talk	PCM16 24 kHz mic → single-use token WSS to agents.assemblyai.com
2 Act	Agent calls dispatch_tool → prefs, conversation, metrics, session
3 Hear	Short spoken reply + live transcript + screen-reader announcements
4 Keep working	Fallback ladder: AssemblyAI → Vosk → Web Speech → typed input

LIVE DEMO

Golden path judges can click in under two minutes

- Sign in → tap Power (or Space) → engine chip: AssemblyAI · healthy
- Open metrics — panel navigation with spoken confirmation
- Speak faster — agent writes preferences via dispatch_tool
- Describe the page — audible DOM/visual summary
- How did I do today? — spoken session metrics from the DB
- Kill the network → typed commands still work (fallback ladder)

DEMO CHECKLIST

Power connects
Voice intent navigates
Tool writes prefs
Metrics speak
Offline fallback
Alt+1-7 shortcuts

TECHNOLOGY

Why AssemblyAI Voice Agent — not generic STT

Voice Agent WS	Full-duplex listen + speak on one socket
Stored agent_id	Prompt, voice, keyterms, tools configured once
Single-use tokens	API key never leaves the server
Function tools	dispatch_tool → tool.call / tool.result loop
Interruption / VAD	Barge-in flushes local playback cleanly
Session lifecycle	session.end + auto-resume in the 30s window

ARCHITECTURE

Browser + server + AssemblyAI, fail-closed security

Browser

- Next.js dashboard
- PCM worklet 24 kHz
- Local client actions
- Engine ladder UI

Server

- /api/agent/token
- /api/agent/tools
- /api/stt/fallback
- Prisma + SQLite

AssemblyAI

- Stored Voice Agent
- Realtime WS session
- LLM + TTS + VAD
- tool.call events

Security: signed HttpOnly sessions · CSRF · rate limits · bot/scanner block · AES-256-GCM · audit log

DIFFERENTIATION

Not another chat wrapper on a form

Chat wrapper	No product actions
Pure STT captions	Passive — cannot act
Offline-only STT	Weak LLM & tools
Cloud-only demo	Dies without network
VoiceBridge	Acts + speaks + degrades gracefully

BUSINESS

Who pays, why now, how we charge

- Primary: blind / low-vision / motor-impaired users
- Buyer: enterprise a11y & digital workplace teams
- Why now: WCAG + EN 301 549 enforcement expanding
- Why AI: realtime voice agent with tools that write product state

Personal $12/mo
Browser agent, fallback, export

Team $49/seat
Shared prefs, history, audit

Enterprise Custom
SSO, private deploy, Twilio phone

BUILD STATUS

Production-shaped, not a weekend spike

16

API routes, auth-wrapped

8/8

Orchestrator unit tests

E2E

Playwright golden path

14

Security controls verified

Stored AssemblyAI agent published · Judge demo video + README + submission kit in repo

NEXT

Roadmap and what we need

- Phone channel via Twilio SIP (AssemblyAI telephony path)
- Multi-user auth + enterprise SSO
- Richer vision tools (element targeting, form fill)
- Bluejay simulation scoring for agent quality

THE ASK

Ship VoiceBridge to users who cannot afford to wait for a mouse-first web to catch up.

Prize credits → production AssemblyAI minutes + phone pilot with accessibility testers.

github.com/BROCKUGANDA/voicebridge-ai · AssemblyAI Voice Agent API